

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

**Product Name** Hydrogen chloride, 1M in diethyl ether

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>H27125</b>                                                                               |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 1

Substances/mixtures corrosive to metal

Category 1

#### Health hazards

Acute Oral Toxicity

Category 4

Acute Inhalation Toxicity - Vapors

Category 3

Skin Corrosion/Irritation

Category 1 A

Serious Eye Damage/Eye Irritation

Category 1

Specific target organ toxicity - (single exposure)

Category 3

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements



**Signal Word**

**Danger**

**Hazard Statements**

H224 - Extremely flammable liquid and vapor  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H331 - Toxic if inhaled  
H336 - May cause drowsiness or dizziness  
H290 - May be corrosive to metals

**Precautionary Statements**

**Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates  
May form explosive peroxides  
Repeated exposure may cause skin dryness or cracking

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Ethyl ether       | 60-29-7   | 90-95    |
| Hydrogen chloride | 7647-01-0 | 5-10     |

## Section 4 - First Aid Measures

Description of first aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Inhalation</b>                          | Remove to fresh air. If breathing is difficult, give oxygen. Immediate medical attention is required. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.                                                                                                                         |
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                                                                                                                                                                               |
| <b>Skin Contact</b>                        | Immediate medical attention is required. Wash off immediately with plenty of water for at least 15 minutes.                                                                                                                                                                                                                                                                                                                                     |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.                                                                                                                                                                                                                                                                                                          |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                                |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes burns by all exposure routes. . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated; Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                                                                          |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Water mist may be used to cool closed containers. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Extremely flammable. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. May re-ignite after fire is extinguished. Thermal decomposition can lead to release of irritating gases and vapors. May form explosive peroxides. Corrosive material. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. Pay attention to flashback. Avoid contact with skin, eyes and inhalation of vapors.

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#### Environmental Precautions

See Section 12 for additional Ecological Information.

#### Methods for Containment and Clean Up

Remove all sources of ignition. Provide adequate ventilation. Use spark-proof tools and explosion-proof equipment. Soak up with inert absorbent material. Keep combustibles (wood, paper, oil, etc) away from spilled material. Keep in suitable, closed containers for disposal.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Do not get in eyes, on skin, or on clothing. Do not breathe mist/vapors/spray. Use caution when opening. Contents may develop pressure upon prolonged storage. If peroxide formation is suspected, do not open or move container. Protect from moisture. Protect from light. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Flammables area. Keep away from heat, sparks and flame. Protect from direct sunlight. Keep under nitrogen. May form explosive peroxides on prolonged storage. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Do not store in metal containers. Keep only in the original container. Keep container tightly closed in a dry and well-ventilated place. Refrigerator/flammables.

#### Incompatible Materials

Bases. Strong oxidizing agents.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component         | New Zealand WEL                                                                              | Australia                                                                                    | ACGIH TLV                     | The United Kingdom                                                                                                 |
|-------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Ethyl ether       | TWA: 400 ppm<br>TWA: 1210 mg/m <sup>3</sup><br>STEL: 500 ppm<br>STEL: 1520 mg/m <sup>3</sup> | STEL: 500 ppm<br>STEL: 1520 mg/m <sup>3</sup><br>TWA: 400 ppm<br>TWA: 1210 mg/m <sup>3</sup> | TWA: 400 ppm<br>STEL: 500 ppm | STEL: 200 ppm 15 min<br>STEL: 620 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 310 mg/m <sup>3</sup> 8 hr |
| Hydrogen chloride | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup>                                             | 7.5mg/m <sup>3</sup> TWA & Peak Limitation                                                   | Ceiling: 2 ppm                | STEL: 5 ppm 15 min<br>STEL: 8 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 2 mg/m <sup>3</sup> 8 hr         |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

|                        |                                                                                                    |
|------------------------|----------------------------------------------------------------------------------------------------|
| <b>Eye Protection</b>  | Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications) |
| <b>Hand Protection</b> | Protective gloves                                                                                  |

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene, Natural rubber, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                 |                                                                                                                                                                                                                                                                                                                             |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>   | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b> | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>  | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                                                |                          |                                          |
|------------------------------------------------|--------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                   |                                          |
| <b>Appearance</b>                              | Colorless                |                                          |
| <b>Odor</b>                                    | No information available |                                          |
| <b>Odor Threshold</b>                          | No data available        |                                          |
| <b>pH</b>                                      | No information available |                                          |
| <b>Melting Point/Range</b>                     | No data available        |                                          |
| <b>Softening Point</b>                         | No data available        |                                          |
| <b>Boiling Point/Range</b>                     | No information available |                                          |
| <b>Flammability (liquid)</b>                   | Extremely flammable      | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available        |                                          |
| <b>Flash Point</b>                             | -40 °C / -40 °F          | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | No data available        |                                          |
| <b>Decomposition Temperature</b>               | No data available        |                                          |
| <b>Viscosity</b>                               | No data available        |                                          |
| <b>Water Solubility</b>                        | Soluble                  |                                          |
| <b>Solubility in other solvents</b>            | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                          |
| <b>Component</b>                               | <b>log Pow</b>           |                                          |
| Ethyl ether                                    | 0.82                     |                                          |
| <b>Vapor Pressure</b>                          | No data available        |                                          |
| <b>Density / Specific Gravity</b>              | 0.731-0.747              |                                          |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                   |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)  |                                          |

**Other information**

**Explosive Properties** Vapors may form explosive mixtures with air

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                                                                                                                                                                   |
| <b>Stability</b>                        | May form explosive peroxides. Reacts with air to form peroxides. Hygroscopic. Light sensitive. Air sensitive.                                                                                                                |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                                     |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                                                                     |
| <b>Hazardous Reactions</b>              | No information available.                                                                                                                                                                                                    |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition, Do not subject to grinding/shock/friction, Exposure to light, Exposure to moist air or water, Incompatible products, Do not distill or allow to evaporate. |
| <b>Incompatible Materials</b>           | Bases, Strong oxidizing agents.                                                                                                                                                                                              |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Hydrogen chloride gas.                                                                                                                                              |

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure**

**Product Information**

|                   |                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.                                                              |
| <b>Eyes</b>       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. |
| <b>Skin</b>       | Avoid contact with skin. Causes burns. Skin Corrosion/Irritation.                               |
| <b>Ingestion</b>  | May be harmful if swallowed.                                                                    |

**Numerical measures of toxicity**

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Category 4                                                       |
| <b>Dermal</b>              | Based on available data, the classification criteria are not met |
| <b>Inhalation</b>          | Based on available data, the classification criteria are not met |

**Toxicology data for the components**

| Component         | LD50 Oral            | LD50 Dermal             | LC50 Inhalation                                                                                                                   |
|-------------------|----------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Ethyl ether       | 1215 mg/kg (Rat)     | 20 mL/kg (Rabbit)       | 32000 ppm ( Rat ) 4 h                                                                                                             |
| Hydrogen chloride | 900 mg/kg ( Rabbit ) | > 5010 mg/kg ( Rabbit ) | LC50 = 4701 ppm (rat) 30 min (gas), LC50 = 588 ppm (4h) by extrapolation<br>LC50 = 8.3 mg/L (rat ) 30 min (aerosols) (MMAD < 5µm) |

|                                               |                                                                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>(b) skin corrosion/irritation;</b>         | Category 1 A                                                                                                      |
| <b>(c) serious eye damage/irritation;</b>     | Category 1                                                                                                        |
| <b>(d) respiratory or skin sensitization;</b> |                                                                                                                   |
| <b>Respiratory</b>                            | Based on available data, the classification criteria are not met                                                  |
| <b>Skin</b>                                   | Based on available data, the classification criteria are not met                                                  |
| <b>(e) germ cell mutagenicity;</b>            | Based on available data, the classification criteria are not met                                                  |
|                                               | Mutagenic effects have occurred in humans                                                                         |
| <b>(f) carcinogenicity;</b>                   | Based on available data, the classification criteria are not met                                                  |
|                                               | There are no known carcinogenic chemicals in this product                                                         |
| <b>(g) reproductive toxicity;</b>             | Based on available data, the classification criteria are not met                                                  |
| <b>(h) STOT-single exposure;</b>              | Category 3                                                                                                        |
| <b>Results / Target organs</b>                | Central nervous system (CNS)                                                                                      |
| <b>(i) STOT-repeated exposure;</b>            | Based on available data, the classification criteria are not met                                                  |
| <b>Target Organs</b>                          | None known.                                                                                                       |
| <b>(j) aspiration hazard;</b>                 | Based on available data, the classification criteria are not met                                                  |
| <b>Other Adverse Effects</b>                  | The toxicological properties have not been fully investigated. See actual entry in RTECS for complete information |

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

## Section 12 - Ecological Information

**Ecotoxicity****Aquatic ecotoxicity**

Do not empty into drains.

| Component   | Freshwater Fish                                                                                                   | Water Flea          | Freshwater Algae | Microtox                |
|-------------|-------------------------------------------------------------------------------------------------------------------|---------------------|------------------|-------------------------|
| Ethyl ether | LC50: > 10000 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 2560 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 165 mg/L/24h |                  | EC50 = 5600 mg/L 15 min |

**Terrestrial ecotoxicity**

There is no data for this product

**Persistence and Degradability****Persistence**

Soluble in water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Ethyl ether | 0.82    | No data available             |

**Mobility**

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**Other adverse effects****Endocrine Disruptor Information  
Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods****Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled



or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

## Section 14 - Transport Information

| Component                               | Hazchem Code |
|-----------------------------------------|--------------|
| Ethyl ether<br>60-29-7 ( 90-95 )        | 3YE          |
| Hydrogen chloride<br>7647-01-0 ( 5-10 ) | 2RE<br>2R    |

### NZS 5433:2020

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2924                              |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s. |
| Hazard Class            | 3                                   |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | I                                   |

### IATA

|                         |                                      |
|-------------------------|--------------------------------------|
| UN-No                   | UN2924                               |
| Proper Shipping Name    | FLAMMABLE LIQUID, CORROSIVE, N.O.S.* |
| Hazard Class            | 3                                    |
| Subsidiary Hazard Class | 8                                    |
| Packing Group           | I                                    |

### IMDG/IMO

|                         |                                     |
|-------------------------|-------------------------------------|
| UN-No                   | UN2924                              |
| Proper Shipping Name    | Flammable liquid, corrosive, n.o.s. |
| Hazard Class            | 3                                   |
| Subsidiary Hazard Class | 8                                   |
| Packing Group           | I                                   |

Environmental hazards No hazards identified

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code Not applicable, packaged goods

Special Precautions No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

| Component         | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | IMDG Marine Pollutant |
|-------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------|
| Hydrogen chloride | 25 tonne                                                                                  | 250 tonne                                                                                |                       |

**Authorisation/Restrictions according to EU REACH**

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrogen chloride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No    | NZIoC | AICS | EINECS | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------|-----------|-------|------|--------|--------|-----|----------|-------|------|
| Ethyl ether       | 60-29-7   | X     | X    | -      | -      | -   | KE-27690 | X     | X    |
| Hydrogen chloride | 7647-01-0 | X     | X    | -      | -      | -   | KE-20189 | X     | X    |

| Component         | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | PICCS | ISHL | ENCS |
|-------------------|-----------|------|-----------------------------------------------|-----|------|-------|------|------|
| Ethyl ether       | 60-29-7   | X    | ACTIVE                                        | X   | -    | X     | X    | X    |
| Hydrogen chloride | 7647-01-0 | X    | ACTIVE                                        | X   | -    | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>IARC</b> - International Agency for Research on Cancer<br><b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land<br><b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association<br><b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships<br><b>LD50</b> - Lethal Dose 50%<br><b>EC50</b> - Effective Concentration 50%<br><b>WEL</b> - Workplace Exposure Limit<br><b>DNEL</b> - Derived No Effect Level<br><b>POW</b> - Partition coefficient Octanol:Water<br><b>vPvB</b> - very Persistent, very Bioaccumulative<br><b>VOC</b> - (Volatile Organic Compound) | <b>PNEC</b> - Predicted No Effect Concentration<br><b>OECD</b> - Organisation for Economic Co-operation and Development<br><b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code<br><b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail<br><b>LC50</b> - Lethal Concentration 50%<br><b>ATE</b> - Acute Toxicity Estimate<br><b>RPE</b> - Respiratory Protective Equipment<br><b>NOEC</b> - No Observed Effect Concentration<br><b>BCF</b> - Bioconcentration factor<br><b>PBT</b> - Persistent, Bioaccumulative, Toxic |
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#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

|                              |                       |
|------------------------------|-----------------------|
| <b>Physical hazards</b>      | On basis of test data |
| <b>Health Hazards</b>        | Calculation method    |
| <b>Environmental hazards</b> | Calculation method    |

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.

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|-------------------------|----------------|
| <b>Revision Date</b>    | 20-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

#### Disclaimer

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End of Safety Data Sheet